

Cross-Border Flow & Torkham Analytics

- **The Gateway:** Torkham is a critical artery for trade. Analyzing "Turnaround Time" (TAT) for trucks helps identify bureaucratic or physical bottlenecks.
- **Geospatial Challenges:** Mountainous terrain (like the Khyber Pass) introduces non-linear variables. A 10km stretch in the plains is not equivalent to 10km of steep, winding elevation in terms of fuel or time
- **Real-Time Tracking:** Using GPS and RFID data to visualize the supply chain pipeline and predict delays before they cascade.



Demand Forecasting (Karkhano Inventory)

- **The Retail Pulse:** Shops in Karkhano Market deal with diverse, high-velocity inventory. Overstocking wastes capital; understocking loses sales.
- **Time-Series Components:**
 - **Trend:** Is a specific electronic brand growing in popularity?
 - **Seasonality:** Do sales spike during wedding seasons or before religious holidays?
- **Objective:** Moving from "Intuition" to "Data-Driven Procurement."



Transport Efficiency & Fleet Management



The Indus Highway: A major corridor where small efficiency gains lead to massive savings.



Fuel Consumption Modeling: Fuel usage is a function of load, speed, and road gradient.



The Efficiency Formula: We calculate the "Cost per Ton-Mile" to benchmark different transport providers.